

OceanEmbed — System Mathematics & Data Blueprint

AI-Powered Ocean Heat Risk Intelligence Layer for Bay of Bengal & Indo-Pacific Maritime Climate Resilience

Domain: Bay of Bengal (5°N–22°N, 80°E–95°E) Grid: 0.25° Daily • Cell-Centered: 68 Lat × 60 Lon cells (~28 km)

Target Depths: 6 Standard Levels: $z_k \in \{0, 50, 100, 200, 500, 1000\}$ m Season: Monsoon Jun 1 – Aug 31, 2026 (92 Days)

1. Executive Summary & Scientific Positioning

Subsurface ocean thermal structure governs upper-ocean heat content, thermocline dynamics, barrier-layer stratification, and tropical cyclone energetics. Drifting ARGO profiling floats and moored buoys provide high-fidelity in-situ profiles but remain sparse in space and time. Conversely, operational satellite and reanalysis constellations provide dense, daily surface observations (SST, altimetric SSH, surface currents, surface winds, and sea surface salinity).

OceanEmbed is a physics-regularized deep learning framework engineered to reconstruct daily depth-resolved 3D ocean temperature fields across 6 standardized levels directly from multi-source satellite and reanalysis input fields over the Bay of Bengal, providing an uncertainty-aware ocean heat intelligence layer for maritime disaster preparedness and climate resilience.

"We do not claim to observe the deep ocean directly from satellites. We quantify how much of the subsurface thermal structure can be reconstructed from multi-source surface signals, compare against simple and strong baselines, constrain predictions with physically motivated regularization, validate separately against a numerical reanalysis target and in-situ observations, and expose uncertainty and limitations rather than hiding them."

2. Comprehensive Data Extraction Contract & Sources

To eliminate ambiguity and prevent unannounced data substitutions, exactly one primary source product is frozen per channel:

Variable	Frozen Source Product	Data Provider & API Source	Native Res.	Regridded Target
Sea Surface Temp. (SST)	OSTIA L4 Foundation SST Analysis	UK Met Office / CMEMS SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001	0.05° daily	0.25° daily mean (5×5 block avg) Core-6 MVP
Sea Level Anomaly (SLA / SSH)	CMEMS DUACS Altimetry Gridded SLA	CNES / CLS / Copernicus Marine Service SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046	0.25° daily	0.25° regular grid alignment Core-6 MVP
Zonal Current (U_cur)	OSCAR v2.0 Final L4 Currents	NASA PO.DAAC / ESR OSCAR_L4_OC_FINAL_V2.0 via PO.DAAC Drive	0.25° daily	0.25° regular grid alignment Core-6 MVP
Meridional Current (V_cur)	OSCAR v2.0 Final L4 Currents	NASA PO.DAAC / ESR OSCAR_L4_OC_FINAL_V2.0 via PO.DAAC Drive	0.25° daily	0.25° regular grid alignment Core-6 MVP
Zonal 10m Wind (U_wind)	ERA5 Atmospheric Reanalysis	ECMWF / Copernicus Climate Change Service (C3S) reanalysis-era5-single-levels via cdsapi	0.25° hourly	0.25° 24h vector daily mean Core-6 MVP
Meridional Wind (V_wind)	ERA5 Atmospheric Reanalysis	ECMWF / Copernicus Climate Change Service (C3S) reanalysis-era5-single-levels via cdsapi	0.25° hourly	0.25° 24h vector daily mean Core-6 MVP
Sea Surface Salinity (SSS)	SMAP L4 / CMEMS Multi-Obs L4	NASA JPL / CMEMS Multi-Obs MULTIOBS_GLO_PHY_SSS_L4_MYNRT_015_013	0.125° daily	0.25° daily mean Core-7 Ablation Track
GLORYS Training Target	GLORYS12V1 Ocean Physics Reanalysis	Mercator Ocean / Copernicus Marine Service GLOBAL_MULTIYEAR_PHY_001_030 (thetao)	1/12° (~8 km) 50 vertical levels	0.25° grid at 6 standard levels: [0, 50, 100, 200, 500, 1000] m
In-Situ Validation Data	ARGO Profiling Floats (CTD Sensors)	INCOIS Argo Data Center / Coriolis GDAC Strict QC (Flag = 1 or 2, good measurements)	Point vertical CTD profiles	Bilinear match at float lat/lon; 100% withheld from model training

Grid Registration Specification (68 × 60): The domain bounds [5°N–22°N, 80°E–95°E] span $\Delta\Phi = 17^\circ$ and $\Delta\Lambda = 15^\circ$. Under standard operational **cell-centered (pixel-is-area) registration** at 0.25° spacing, coordinates represent cell centers: Lat $\in [5.125^\circ, 5.375^\circ, \dots, 21.875^\circ\text{N}]$ (exactly $17 / 0.25 = 68$ cells) and Lon $\in [80.125^\circ, 80.375^\circ, \dots, 94.875^\circ\text{E}]$ (exactly $15 / 0.25 = 60$ cells). If registered on node boundaries, dimensions are 69×61 . OceanEmbed freezes **68 × 60 cell-centered registration** matching CMEMS L4 standards.

Operational Data Fallback (Core-6-Lite): If upstream latency or authentication issues occur with OSCAR currents during initial ingestion, the documented fallback model Core-6-Lite operates strictly on [SST, SLA, U_wind, V_wind]. Core-6-Lite is an operational continuity fallback and is evaluated on aligned stencils.

Scientific Transparency Disclosures: 1. *ERA5 surface winds* are atmospheric reanalysis fields rather than raw scatterometer observations; they serve as standard operational proxies for surface wind stress curl ($\nabla \times \tau$). 2. *GLORYS12V1* is an assimilative numerical ocean reanalysis model acting as our dense spatial training target, not absolute ground truth. Real-world observational fidelity is quantified via independent ARGO matchup benchmarks. 3. *Spatial Regridding & Partitioning Discipline:* All spatial regridding is performed before chronological partitioning for grid consistency; all statistical parameters (μ, σ) are fitted strictly on training dates. Provenance, product versions, and units are serialized to metadata.json.

3. Mathematical Anomaly Formulation & Climatological Prior

Directly predicting raw subsurface temperatures with neural networks causes severe optimization bias: the model easily memorizes the large vertical stratification gradient ($\sim 28^\circ\text{C}$ at surface to $\sim 5^\circ\text{C}$ at 1000m) while missing dynamic thermocline displacements caused by mesoscale eddies and wind-driven upwelling.

To resolve this, OceanEmbed models temperature as an **anomaly reconstruction** on top of a **spatially varying, training-safe monthly climatological background**:

EQUATION 1: FULLY INDEXED SUBSURFACE RECONSTRUCTION FORMULATION

$$\hat{T}_{b,k,x,y,t} = \bar{T}_{\text{clim}}(k, x, y, m(t)) + \sigma_{\text{train},k} \cdot \Delta \hat{T}_{\text{norm},b,k,x,y,t}$$

for batch index b , depth level index $k \in \{0, 1, 2, 3, 4, 5\}$ (corresponding to depths $z_k \in \{0, 50, 100, 200, 500, 1000\}$ m), spatial coordinates (x, y) on the 68×60 grid, day t , and month $m(t) \in \{\text{June}, \text{July}\}$. $\bar{T}_{\text{clim}} \in \mathbb{R}^{[B \times 1 \times 68 \times 60]}$ is broadcast spatially per depth level, capturing horizontal oceanographic gradients across the Bay.

EQUATION 2: TARGET STANDARDIZATION RULE WITH VALID-SAMPLE DENOMINATOR

$$\Delta T_{\text{norm},k}(t, x, y) = \frac{T_{\text{GLORYS}}(t, k, x, y) - \bar{T}_{\text{clim}}(k, x, y, m(t))}{\sigma_{\text{train},k}}$$

where the depth-wise scalar standard deviation $\sigma_{\text{train},k}$ is computed strictly over the training partition with dynamic valid mask M_{txy} :

$$\sigma_{\text{train},k} = \sqrt{\text{Big}(\frac{\sum_{t \in T_{\text{train}}} \sum_{x,y} M_{\text{txy}} \cdot [T_{\text{GLORYS}}(t, k, x, y) - \bar{T}_{\text{clim}}(k, x, y, m(t))]^2}{\sum_{t \in T_{\text{train}}} \sum_{x,y} M_{\text{txy}} + \varepsilon}) \text{Big})}$$

Chronological Splitting Protocol & Data Leakage Shield

Training Set: June 1 – July 31, 2026 (61 daily timesteps) — Covers southwest monsoon onset and development.

Validation Set: August 1 – August 15, 2026 (15 daily timesteps) — Model checkpointing, early stopping, and hyperparameter tuning.

Held-Out Test Block: August 16 – August 31, 2026 (16 daily timesteps) — Strictly frozen evaluation benchmark.

Climatological Prior Shield: $\bar{T}_{\text{clim}}$ is derived strictly from June–July 2026 and historical GLORYS baselines. The test period (August 16–31) is strictly excluded to prevent temporal information leakage.

Common-Support Evaluation Rule: Core-6 versus Core-7 ablation metrics are computed on their common valid spatiotemporal support ($T_{\text{test}} \cap M_{\text{common}}$), ensuring performance differences are attributable to SSS rather than missing-data masks.

4. Core Neural Architecture: ResNet-18 Encoder-Decoder with Target-Size Alignment

To preserve the full 2D spatial grid (68×60) without patch-stitching artifacts, OceanEmbed uses an end-to-end Fully Convolutional network with explicit depth conditioning and bilinear target-size interpolation:

INPUT SURFACE TENSOR: $X \in \mathbb{R}^{[\text{Batch} \times C_{\text{in}} \times 68 \times 60]}$ ($C_{\text{in}}=6$ for Core-6: [SST, SLA, U_{cur} , V_{cur} , U_{wind} , V_{wind}])

ResNet-18 BACKBONE ENCODER:

- First conv adapted from $3 \rightarrow C_{\text{in}}$ channels: $W_{\text{init}}^{(c)} = W_{\text{RGB}}^{(c)}$ for $c \leq 3$; $W_{\text{init}}^{(c)} = (\frac{1}{3}) \sum W_{\text{RGB}} \cdot \sqrt{3 / C_{\text{in}}}$ for $c > 3$
- Spatial downsampling by factor 8 (output stride 8): Conv1+MaxPool (34×30), Layer 1 (34×30), Layer 2 (17×15), Layer 3 (9×8)
- Bottleneck Feature Map:** $F_{\text{enc}} \in \mathbb{R}^{[\text{Batch} \times 128 \times 9 \times 8]}$

DECODER WITH TARGET-SIZE BILINEAR RESTORATION:

- 3 Upsampling stages with lateral skip connections from Layer 2 (17×15) and Layer 1 (34×30)
- Exact Dimension Reconciliation:** Rather than naive $2 \times$ scaling (which yields 72×64 from 9×8), each decoder stage aligns to skip dimensions via `interpolate(size=skip.shape[-2:], mode='bilinear')`, and final stage interpolates to **exact input grid (68×60)**.
- Restored Spatial Latent Field:** $F \in \mathbb{R}^{[\text{Batch} \times 64 \times 68 \times 60]}$

DEPTH-CONDITIONED 1×1 CONVOLUTIONAL PROJECTION HEAD:

- 6 Learned depth embedding vectors: $d_k \in \mathbb{R}^{16}$ broadcast across spatial grid (16 channels)
- Spatially broadcast climatology prior: $T_{\text{clim}}(k, x, y) \in \mathbb{R}^{[\text{Batch} \times 1 \times 68 \times 60]}$ (1 channel)
- Concatenated Representation:** $\Phi_k = [F \oplus d_k \oplus T_{\text{clim}}(k)] \in \mathbb{R}^{[\text{Batch} \times 81 \times 68 \times 60]}$ (81 channels total)
- 1×1 Convolutions: $81 \rightarrow 64$ (GELU) $\rightarrow 32$ (GELU) $\rightarrow 1$ (Linear Normalized Anomaly $\Delta \hat{T}_{\text{norm}}$) per depth level

OUTPUT RECONSTRUCTED 3D THERMAL FIELD: $\hat{T} \in \mathbb{R}^{[\text{Batch} \times 6 \times 68 \times 60]}$ (Standard Depths: 0, 50, 100, 200, 500, 1000 m)

- Monte Carlo Dropout Inference Protocol:** Model is set to `model.eval()` (freezing BatchNorm running statistics); only nn.Dropout layers are dynamically activated ($p=0.1$) for $S=10$ forward passes:

$$\text{Predictive Mean: } \bar{T}_{\text{MC}} = \frac{1}{S} \sum \hat{T}^{(s)} \quad | \quad \text{Epistemic Uncertainty: } \sigma_{\text{epistemic}} = \sqrt{\frac{1}{S-1} \sum (\hat{T}^{(s)} - \bar{T}_{\text{MC}})^2}$$

Dashboard Caveat: "MC-dropout epistemic uncertainty: model sensitivity to learned parameters; not total observational uncertainty."

5. Loss Function Formulation & Dimensional Scaling

All temperature predictions and targets are de-standardized to physical Celsius before loss computation. The reconstruction, surface-anchor, and curvature terms have different derived units; λ_{surf} and λ_{smooth} are validation-calibrated balancing coefficients that render the composite optimization objective numerically comparable over the valid-ocean mask $M_{b,k,x,y} \in \{0, 1\}$:

EQUATION 3: SCALE-CALIBRATED COMPOSITE OBJECTIVE FUNCTION (OPTION A FORMULATION)

$$\mathcal{L}_{core} = \mathcal{L}_{weighted_MSE} + \lambda_{surf} \cdot \mathcal{L}_{surf} + \lambda_{smooth} \cdot \mathcal{L}_{curvature}$$

Dimensional Interpretation & Calibration: $\mathcal{L}_{weighted_MSE}$ is in $(^\circ\text{C})^2$ (typical magnitude $\sim 0.3\text{--}0.8$); $\mathcal{L}_{surf}$ is an L_1 MAE in $^\circ\text{C}$ (typical magnitude $\sim 0.2\text{--}0.4$); $\mathcal{L}_{curvature}$ is in $(^\circ\text{C} / (100\text{ m})^2)^2$ (typical magnitude $\sim 1.0\text{--}2.5$). The coefficients λ_{surf} (unit: $^\circ\text{C}$) and λ_{smooth} (unit: $(100\text{ m})^4 / ^\circ\text{C}^2$) are empirically calibrated balancing hyperparameters selected via validation-set loss monitoring and held fixed during test evaluation (nominal starting values: $\lambda_{surf} = 0.5$, $\lambda_{smooth} = 0.05\text{--}0.10$).

5.1 Masked Normalized Depth-Weighted Reconstruction Loss ($\mathcal{L}_{weighted_MSE}$)

EQUATION 4: DEPTH-WEIGHTED RECONSTRUCTION LOSS WITH INDEPENDENT DENOMINATORS

$$\mathcal{L}_{weighted_MSE} = \sum_{k=0}^5 \tilde{w}_k \cdot \frac{\sum_{b,x,y} M_{b,k,x,y} \cdot [\hat{T}_{b,k,x,y} - T_{GLORYS,b,k,x,y}]^2}{\sum_{b,x,y} M_{b,k,x,y} + \varepsilon}$$

Raw thermocline weights: $w = [1.0, 3.0, 3.0, 2.0, 1.0, 0.5]$ for depths $[0, 50, 100, 200, 500, 1000]\text{ m}$.

Mean-normalized weights: $\tilde{w}_k = w_k / \text{mean}(w) = w_k / 1.75 \Rightarrow [0.5714, 1.7143, 1.7143, 1.1429, 0.5714, 0.2857]$. Independent denominators per depth guarantee exact normalization if data coverage varies with depth.

5.2 Soft Surface Constraint Regularizer & Training SST Variance Protocol

EQUATION 5: SOFT SATELLITE SST ANCHORING REGULARIZER (ZERO-INDEXED SURFACE $Z_0 = 0\text{ M}$)

$$\mathcal{L}_{surf} = \frac{\sum_{b,x,y} M_{b,0,x,y} \cdot |\hat{T}_{b,0,x,y} - SST_{OSTIA,b,x,y}|}{\sum_{b,x,y} M_{b,0,x,y} + \varepsilon}$$

Training SST Dispersion Protocol: The training-set SST standard deviation is evaluated strictly from training data:

$$\mu_{SST,train} = \sum_{txy} M_{txy}^{SST} SST_{txy} / (\sum_{txy} M_{txy}^{SST} + \varepsilon) \quad | \quad \sigma_{SST,train} = \sqrt{[\sum_{t \in T_{train},xy} M_{txy}^{SST} (SST_{txy} - \mu_{SST,train})^2] / (\sum M_{txy}^{SST} + \varepsilon)}$$

The actual value is logged in experiment metadata; $\sim 1.25^\circ\text{C}$ is an illustrative empirical expectation, not a predeclared constant.

5.3 Masked Dimensionally Scaled Non-Uniform Vertical Curvature Loss ($\mathcal{L}_{curvature}$)

Rescaling depths by $z_{scale} = 100.0\text{ m}$ gives $z' = [0.0, 0.5, 1.0, 2.0, 5.0, 10.0]$. For internal depth nodes $k \in \{1, 2, 3, 4\}$ (corresponding to 50, 100, 200, 500 m):

EQUATION 6: DIMENSIONALLY SCALED NON-UNIFORM SECOND-ORDER VERTICAL DIFFERENCE

$$\begin{aligned} h_k^- &= z'_k - z'_{k-1}, & h_k^+ &= z'_{k+1} - z'_k \\ \nabla_z^- \hat{T}_k &= \frac{\hat{T}_k - \hat{T}_{k-1}}{h_k^-}, & \nabla_z^+ \hat{T}_k &= \frac{\hat{T}_{k+1} - \hat{T}_k}{h_k^+} \Rightarrow \kappa_k(b, x, y) = 2.0 \cdot \frac{\nabla_z^+ \hat{T}_k - \nabla_z^- \hat{T}_k}{h_k^- + h_k^+} \\ \mathcal{L}_{curvature} &= \frac{\sum_{b,k,x,y} M_{curv,b,k,x,y} \cdot [\kappa_k(b, x, y)]^2}{\sum_{b,k,x,y} M_{curv,b,k,x,y} + \varepsilon} \quad \text{where } M_{curv,b,k,x,y} = M_{b,k-1,x,y} \cdot M_{b,k,x,y} \cdot M_{b,k+1,x,y} \text{ (3-level ocean stencil intersection)} \end{aligned}$$

Physical Interpretation: κ_k has units of $^\circ\text{C} / (100\text{ m})^2$. It penalizes unphysical numerical oscillations across layers while preserving natural stratification.

PyTorch Implementation Blueprint:

```
def nonuniform_vertical_curvature_loss(t_pred, depths_m, mask=None, z_scale=100.0):
    # t_pred: [B, 6, H, W] in Celsius | depths_m: [0, 50, 100, 200, 500, 1000] | mask: [B, 6, H, W]
    z = depths_m.to(t_pred.device, t_pred.dtype) / z_scale
    h_back = (z[1:-1] - z[:-2]).view(1, -1, 1, 1) # z_k - z_{k-1}
    h_forward = (z[2:] - z[1:-1]).view(1, -1, 1, 1) # z_{k+1} - z_k

    grad_back = (t_pred[:, 1:-1] - t_pred[:, :-2]) / h_back
    grad_forward = (t_pred[:, 2:] - t_pred[:, 1:-1]) / h_forward
    curvature = 2.0 * (grad_forward - grad_back) / (h_back + h_forward)
    curv_sq = curvature.square()

    if mask is not None:
        if mask.dim() == 3: mask = mask.unsqueeze(1).expand_as(t_pred)
        # Intersection of 3 participating levels (k-1, k, k+1)
        curv_mask = mask[:, :-2] * mask[:, 1:-1] * mask[:, 2:]
        return (curv_sq * curv_mask).sum() / (curv_mask.sum() + 1e-6)
    return curv_sq.mean()
```

6. Operational Diagnostics: Caveats, Formulations & Edge Cases

Because OceanEmbed reconstructs temperature at 6 discrete standard levels, high-order oceanographic diagnostics are derived via piecewise continuous vertical interpolation. All diagnostic maps carry the mandatory badge:

Dashboard Caveat Badge: “Estimated TCHP / Estimated D26 — Derived from 6-level vertically interpolated profile.”

6.1 Estimated 26°C Isotherm Depth (D26) & Rigorous Edge-Case Rules

ALGORITHM: VERTICAL LINEAR INTERPOLATION OF D26 WITH FULL BOUNDARY HANDLING

- Case 1 (Surface Cold): If $SST \leq 26.0^\circ\text{C} \Rightarrow D26 = 0.0 \text{ m}$.
- Case 2 (Single Standard Crossing): If $SST > 26.0^\circ\text{C}$, locate bracket (z_a, z_b) where $\hat{T}(z_a) \geq 26.0^\circ\text{C}$ and $\hat{T}(z_b) < 26.0^\circ\text{C}$:

$$D26 = z_a + (z_b - z_a) \cdot \frac{\hat{T}(z_a) - 26.0}{\hat{T}(z_a) - \hat{T}(z_b)}$$
- Case 3 (Deep Warmth / Censored): If $\hat{T}(z=1000\text{m}) \geq 26.0^\circ\text{C}$, crossing lies outside domain $\Rightarrow D26 = 1000 \text{ m}$ with metadata flag `censored_below_domain = True`.
- Case 4 (Multiple Crossings / Inversion): In barrier-layer conditions with multiple crossings, D26 is explicitly defined as the shallowest downward crossing (base of the uppermost warm layer available for cyclone energetics) and flagged with `multiple_crossings = True`.
- Case 5 (Surface Inconsistency): If $\hat{T}(z=0\text{m}) < 26.0^\circ\text{C}$ while $SST_{OSTIA} > 26.0^\circ\text{C} \Rightarrow$ Flag `surface_sst_inconsistency = True`.

6.2 Estimated Tropical Cyclone Heat Potential (TCHP) & Censored Lower Bound

EQUATION 7: TROPICAL CYCLONE HEAT POTENTIAL QUADRATURE & UNIT DERIVATION

$$\text{TCHP} = \rho_0 \cdot C_p \cdot \int_0^{D26} \max(0, \hat{T}(z) - 26.0) dz$$

Exact SI to kJ/cm² Conversion: Standard oceanographic constants $\rho_0 = 1025 \text{ kg/m}^3$, $C_p = 3993 \text{ J/(kg}\cdot\text{K)}$ yield volumetric heat capacity:

$\rho_0 C_p = 4,092,825 \text{ J/(m}^3\cdot^\circ\text{C)}$. Since $1 \text{ kJ/cm}^2 = 10^3 \text{ J} / (10^{-2} \text{ m})^2 = 10^7 \text{ J/m}^2$, the conversion factor is:

Factor = $4,092,825 / 10^7 = 0.4092825 \text{ kJ/(cm}^2\cdot^\circ\text{C}\cdot\text{m)}$.

$$\text{TCHP} \approx 0.40928 \cdot \sum_{i=0}^{m-1} \frac{\max(0, T_i - 26) + \max(0, T_{i+1} - 26)}{2} \cdot (z_{i+1} - z_i)$$

with the final segment ending at an explicitly interpolated endpoint: $z_m = D26$ with $T_m = 26.0^\circ\text{C}$.

Censored Domain Rule: When `censored_below_domain = True`, TCHP is designated **TCHP_{0–1000}** (**TCHP_{lower_bound}**) because $\text{TCHP}_{0-1000} \leq \text{TCHP}_{\text{true}}$.

Dashboard Label: “TCHP is a lower bound: D26 lies below the 1000 m reconstruction domain.”

“Cells exceeding 80 kJ/cm^2 are visually flagged as elevated upper-ocean heat-content conditions, a commonly used research indicator relevant to tropical-cyclone intensification. This prototype is an analytical research prototype, not an operational cyclone-warning product.”

6.3 Physical Quality Control & Oceanographic Nuances

Deep Monotonicity Review Rule: At depths $z \geq 300 \text{ m}$, flag positive vertical temperature gradients exceeding **$0.05^\circ\text{C} / 100 \text{ m}$** (i.e. $\Delta T > 0.25^\circ\text{C}$ over the 500–1000m layer) for visual review. Do not automatically reject profiles solely because of an inversion.

Upper-Ocean Inversions (0–100m): Subsurface thermal inversions in the upper 100m are oceanographically valid in the northern Bay of Bengal due to river runoff barrier layers (freshwater lens from Ganges-Brahmaputra).

Linearized Thermosteric Diagnostic: $\eta_{\text{thermo}} = \int_0^{1000\text{m}} \alpha(z) \Delta T(z) dz$ is evaluated post-hoc using depth-dependent thermal expansion $\alpha(z)$ as a weak diagnostic correlation check with altimetric SLA, not as an enforced equality.

7. Independent In-Situ ARGO Collocation Protocol

To evaluate real-world accuracy, independent in-situ ARGO profiling floats are collocated via 4-point geospatial bilinear interpolation:

PROTOCOL: 4-POINT BILINEAR SPATIAL COLLOCATION

$$\hat{T}(\lambda_{\text{argo}}, \varphi_{\text{argo}}, z_k) = (1-u)(1-v) \hat{T}_{i,j} + u(1-v) \hat{T}_{i+1,j} + (1-u)v \hat{T}_{i,j+1} + uv \hat{T}_{i+1,j+1}$$

where $u = (\lambda_{\text{argo}} - \lambda_j) / \Delta\lambda$, $v = (\varphi_{\text{argo}} - \varphi_j) / \Delta\varphi$, with resolution $\Delta\lambda = \Delta\varphi = 0.25^\circ$.

Interpolation & Filtering Protocol: 1. Spatial interpolation: 4-point bilinear on 0.25° grid (all 4 points must be valid ocean).

2. Vertical matching: linear interpolation between the 6 predicted standard levels to Argo CTD observation depths.

3. Observation range: **truncate evaluation to the 0–1000 m interval** (observations deeper than 1000 m are excluded from evaluation; no vertical extrapolation is performed).

4. Temporal matching: nearest daily model snapshot within ± 12 hours.

8. Scientific Benchmark Suite (Populated After Evaluation)

Evaluated strictly on the held-out test block (August 16–31, 2026) over the common valid support ($T_{test} \cap M_{common}$):

Target Depth (z)	Climatology RMSE	SST Linear Regr.	Multi-Source XGBoost	OceanEmbed Core-6	OceanEmbed Core-7 (+SSS)	Independent ARGO Matchups
0 m (Surface SST)	0.6029°C	0.0000°C	0.0032°C	0.4917°C	0.4659°C (SS: +0.4027)	0.312°C (98.9% Acc)
50 m (Upper TC)	0.9227°C	0.5004°C	0.2843°C	0.7744°C	0.7571°C (SS: +0.3266)	0.485°C (98.3% Acc)
100 m (TC Core)	0.5324°C	0.6885°C	0.3245°C	0.5301°C	0.5444°C (SS: +0.0087)	0.542°C (97.8% Acc)
200 m (Lower TC)	0.2840°C	0.3827°C	0.2944°C	0.2829°C	0.2964°C (SS: +0.0081)	0.380°C (98.6% Acc)
500 m (Intermediate)	0.0934°C	0.2660°C	0.2655°C	0.0929°C	0.1011°C (SS: +0.0117)	0.198°C (99.2% Acc)
1000 m (Deep Abyssal)	0.0415°C	0.2521°C	0.2558°C	0.0416°C	0.0455°C (SS: ±0.000)	0.141°C (99.4% Acc)
Full-Profile Metrics	0.4128°C (Prior)	0.3483°C	0.2379°C	0.3689°C	0.3684°C (SS_ov: +0.2532)	0.343°C (98.7% Acc)

SKILL SCORE FORMULATIONS (OVERALL & THERMOCLINE-SPECIFIC)

$$SS_{\text{overall}} = 1 - \frac{\sum_{t \in T_{\text{test}}} \sum_{k=0}^5 \sum_{x,y} M_{t,k,x,y} \cdot (\hat{T} - T_{\text{true}})^2}{\sum_{t \in T_{\text{test}}} \sum_{k=0}^5 \sum_{x,y} M_{t,k,x,y} \cdot (\hat{T}_{\text{clim}} - T_{\text{true}})^2} \quad \bigg| \quad SS_{\text{TC}} = 1 - \frac{\sum_{t \in T_{\text{test}}} \sum_{k \in \{1,2,3\}} \sum_{x,y} M_{t,k,x,y} \cdot (\hat{T} - T_{\text{true}})^2}{\sum_{t \in T_{\text{test}}} \sum_{k \in \{1,2,3\}} \sum_{x,y} M_{t,k,x,y} \cdot (\hat{T}_{\text{clim}} - T_{\text{true}})^2}$$

Reports both unweighted depth-wise skill scores SS_k and the aggregated thermocline skill score SS_{TC}

9. Defensible Jury Defense: Precise Q&A Scripts

Q1: “GLORYS is an assimilative reanalysis, not ground truth. Isn't your ARGO validation compromised by data dependence?”
“Argo profiles are withheld from our machine-learning training and selection pipeline. Because GLORYS is an assimilative reanalysis, Argo is not completely independent of the reanalysis target. We therefore report two distinct evaluations: held-out GLORYS reconstruction skill and direct Argo observational matchups. We do not equate either metric alone with absolute truth.”

Q2: “Does your Core-7 model solve the full halosteric sea level equation?”
“Core-7 tests whether surface salinity improves temperature reconstruction in the Bay of Bengal's freshwater-influenced upper ocean. We do not claim it provides a full subsurface salinity reconstruction or exact halosteric closure. Thermosteric–SLA comparison is retained as a weak diagnostic, not a strict physical equality.”

Q3: “Why did you choose a ResNet-18 rather than a Vision Transformer or FNO?”
“The training set contains 61 temporally distinct daily fields. Although each map contains many spatial cells, those cells are correlated; therefore, large Vision Transformers and Fourier Neural Operators are sample-constrained and susceptible to overfitting. A compact residual CNN provides strong spatial inductive biases, converges rapidly and stably on limited GPU compute, and allows us to focus our engineering on climatological prior injection, non-uniform vertical curvature regularization, and rigorous baseline benchmarking.”

Q4: “Why don't you claim continuous depth reconstruction?”
“Because our training targets are resolved at 6 discrete standard levels, claiming true continuous depth reconstruction would be an overclaim. We interpolate vertically between predicted levels to derive prototype diagnostics like estimated D26 and TCHP, and we explicitly document that interpolation assumption on every plot.”

10. Hackathon Execution Roadmap & Prototype Modules

Phase / Window	Key Deliverables	Core Modules / Responsible
Phase 1 (Hours 00–10)	Download Tier 1 data; regrid to 0.25°; train Climatology, Linear, and XGBoost baselines; lock Phase 1 numbers.	data_loader.py, baselines.py Neelima (Lead), Srihari (Baselines)
Phase 2 (Hours 10–20)	ResNet-18 encoder-decoder model; non-uniform curvature loss; PyTorch Lightning training on Core-6.	model.py, losses.py, train.py Srihari (Model), Ysaswini (Training)
Phase 3 (Hours 20–28)	Core-7 SSS ablation training; bilinear ARGO collocation; benchmark table generation on common support.	argo_collocation.py, benchmark.py Yasaswini (Val), Neelima (Argo)
Phase 4 (Hours 28–34)	FastAPI backend with cached Parquet predictions; React console with diagnostic caveat badges.	api.py, frontend/ Shiva (API), Momin (Dashboard UI)
Phase 5 (Hours 34–36)	Presentation slides finalization; 5-minute pitch rehearsal; jury Q&A dry run; backup demo video.	tests/, run_demo.bat Entire Team (Led by Team Lead)